



FIRST YEAR ELECTRONICS WORKSHOP

Presentation By Alexander Burovanov (President)
IEEE-Eta Kappa Nu (HKN)
Mu Kappa Chapter



Safety & House Keeping

- Electronics are a really fun & awesome! BUT they are a FIRE HAZARD! Please be careful with your batteries especially!
- Any unsafe behaviour will be reported to UQ Security!
- Please make sure you have signed our attendance sheet! This helps IEEE HKN get funding for future events :)





IEEE-Eta Kappa Nu (HKN)



IEEE HKN Mu Kappa Chapter

01

Who Are We?

IEEE Honor Society of Electrical and Computer Engineers, is an honor society for students and professionals in the fields of electrical and computer engineering. It is a part of the Institute of Electrical and Electronics Engineers (IEEE), which is a major international organization dedicated to advancing technology.

02

What We Do?

IEEE HKN Mu Kappa Chapter facilitates technical presentations, workshops and events for the IEEE community here at The University of Queensland.

03

How Can You Get Involved?

If you like the event today and want to get involved, please follow us on Facebook and keep an eye out for future events!

If you are interested in joining IEEE HKN then please reach out to us once you have completed Year 2, Semester 1.





IEEE-Eta Kappa Nu (HKN)



IEEE Who We Are!

- Institute of Electrical and Electronics Engineers (IEEE)
- Largest Globally recognised professional association for Engineering!
- More than 450,000 members in over 190 countries!
- Non-Profit Organization
- There are 39 Societies under IEEE (HKN is one of them!)

Student Membership
(\$13.50 USD)





IEEE In Australia



IEEE Queensland Section

UQ IEEE Sub-Committees

UQ IEEE Student Branch
ge.zhang@uq.edu.au

IEEE HKN Mu Kappa Chapter
a.burovanov@student.uq.edu.au

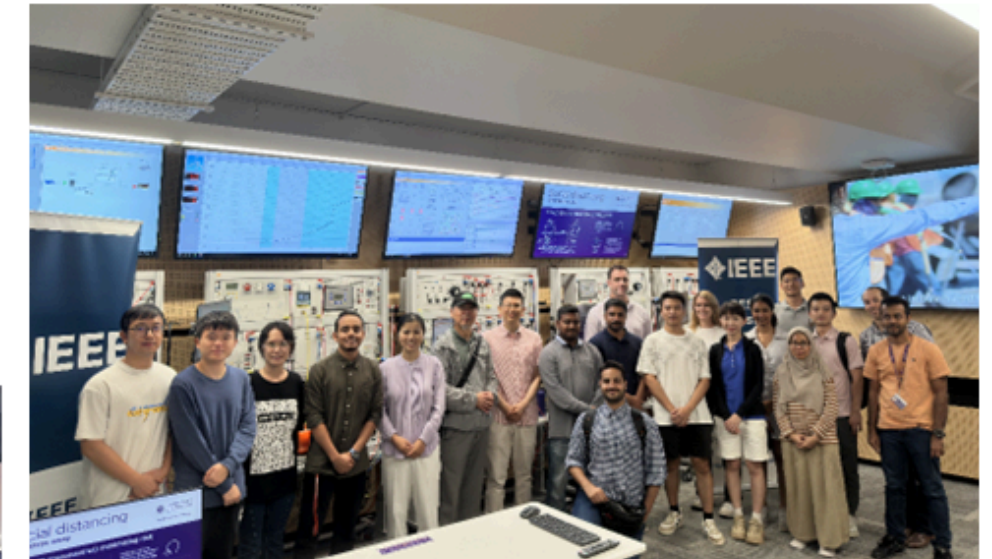
UQ IEEE PES Student Branch
taolue.shen@uq.edu.au



IEEE-Eta Kappa Nu (HKN)



Past Events



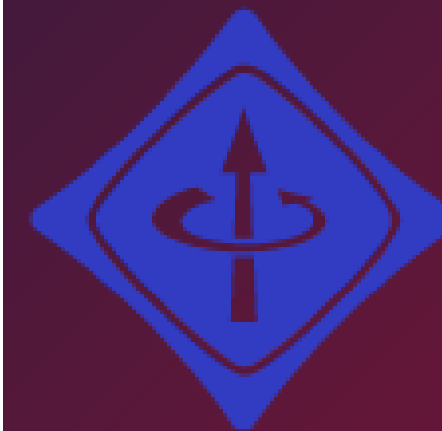


IEEE-Eta Kappa Nu (HKN)

Join the Volunteer Team of IEEE Student Branch!

- More than just participate, work as a planner, organiser!
- Join the most active IEEE Student Organisation in Australia!
- Opportunity to connect with your future supervisor and industry leaders!
- Opportunity to collaborate with and join PES and HKN societies!
- Build up your skills and network!

IEEE Student Membership
(\$13.50 USD)



IEEE

UQ IEEE Student Branch
Volunteer Signup



What is an Arduino?

Open-Source Electronics Prototyping Platform

01

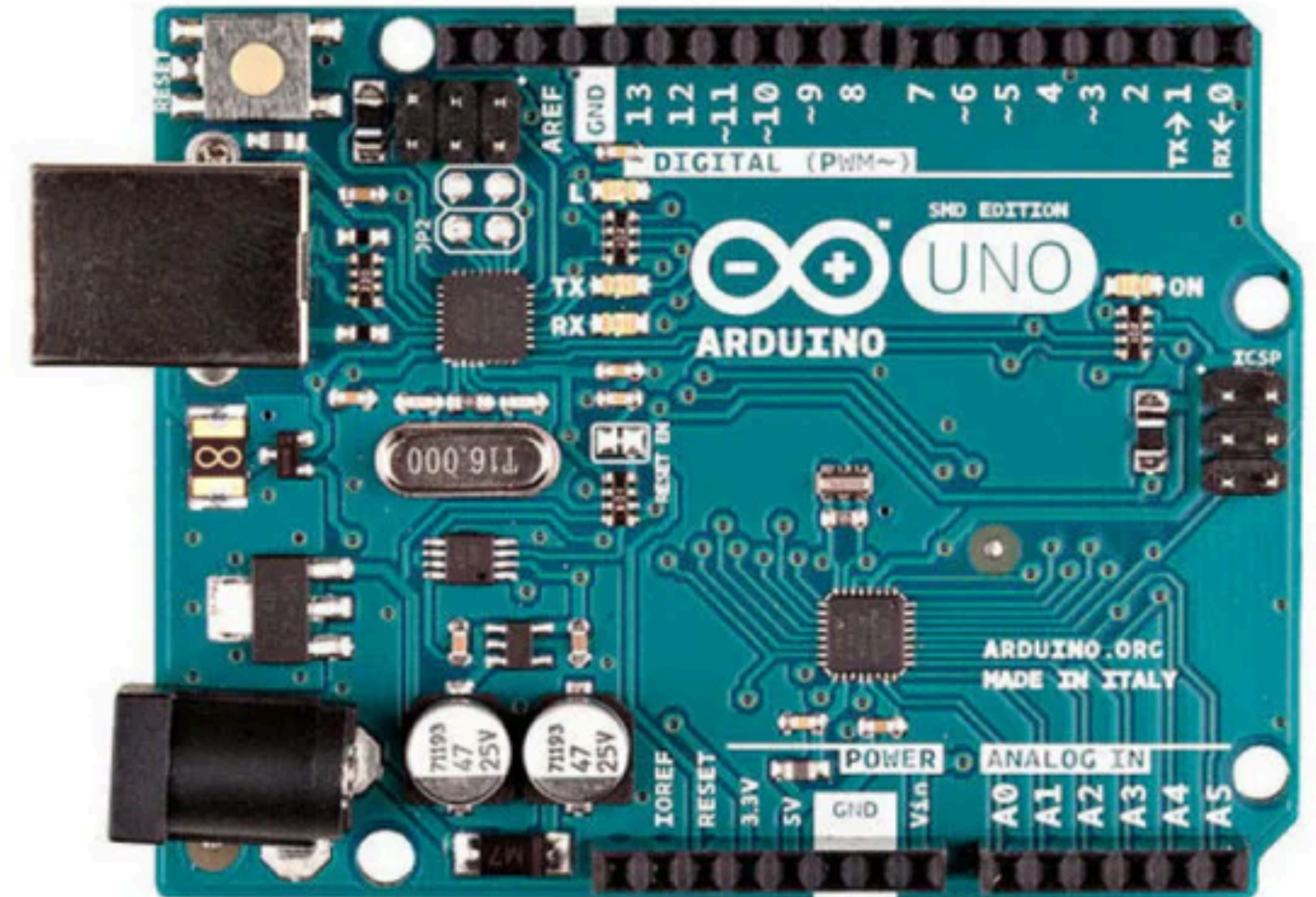
Arduino Hardware

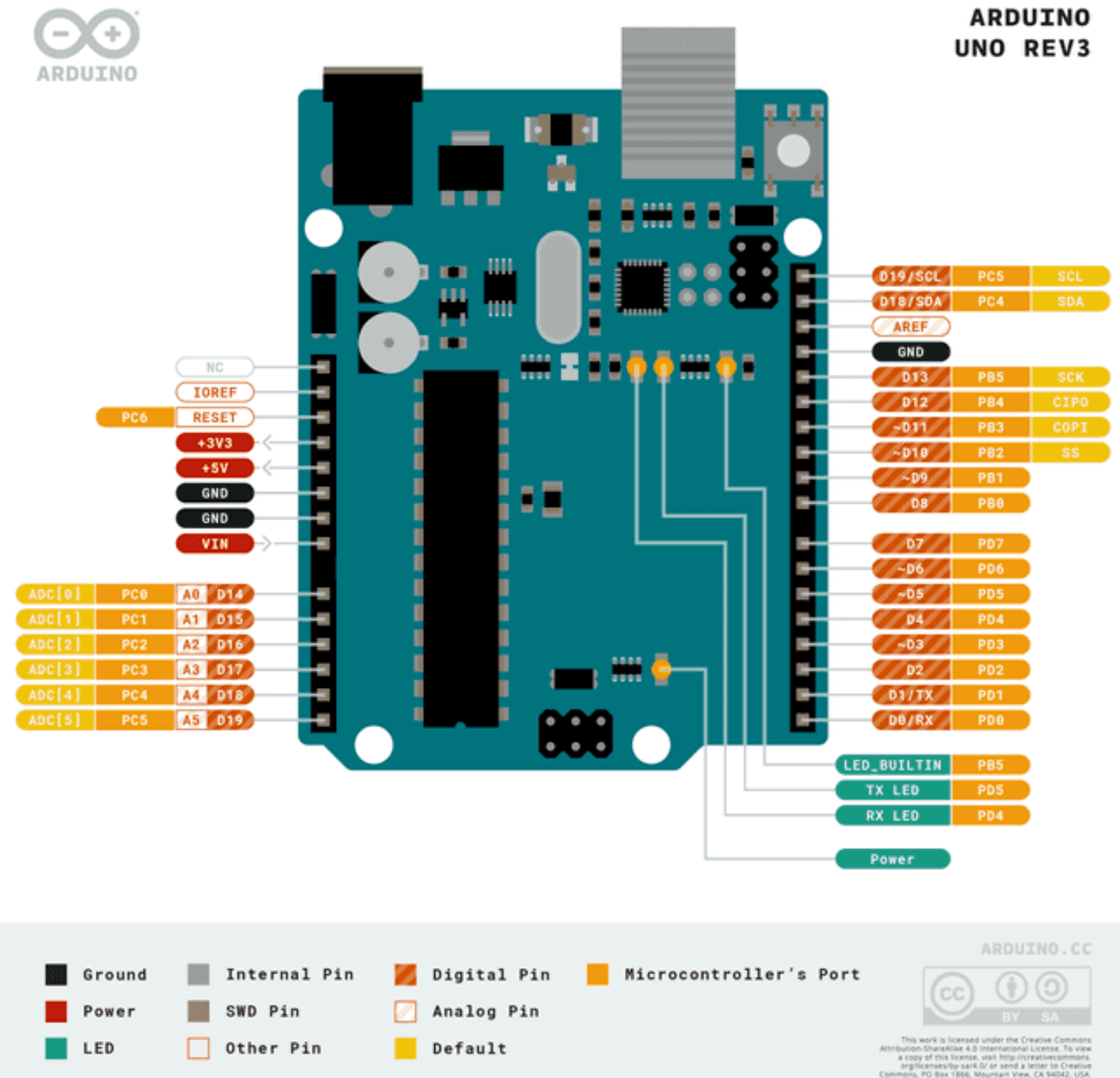
- The Arduino boards are called Development Boards
 - These boards have a range of peripherals on them such as Analog->Digital Converter, LEDs & Push Buttons.
 - The most common Arduino you will see is the Arduino UNO R3! This is a development board with an ATmega328p microcontroller. The Uno R4 is based uses the ESP32 microcontroller.
- Why do we bother with Arduino? Because it is a lot faster and easier than designing and fabricating a circuit board!

02

Arduino IDE

- IDE = Integrated Development Environment
- This is where you write the code that will be stored and executed on your Arduino Board
- The Arduino IDE has a lot of useful features which we will demonstrate today!





Terminology Alert!!!!
GPIO Pins = General Purpose Input Output Pins
You can configure these pins to do anything in code!
PWM Pins = Pulse Width Modulation Pins

Arduino Uno Pinout

The pins are not all the same! Here is a rundown on what the different pins are for.

01

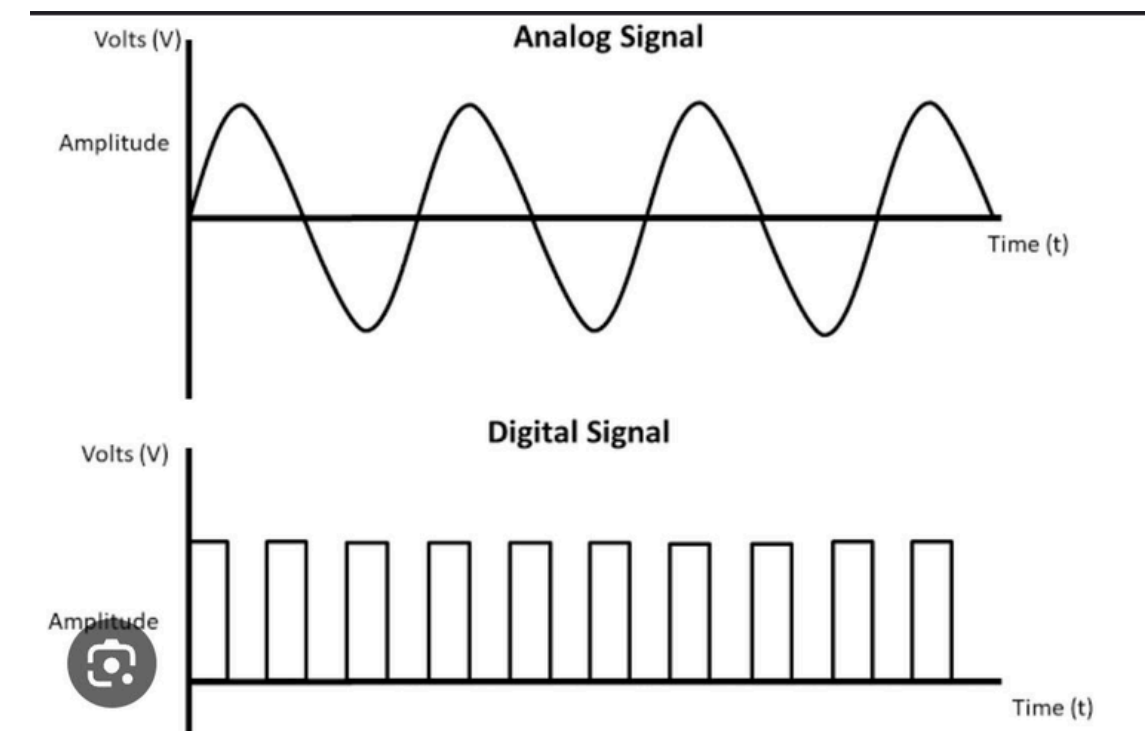
Analog Pins

- Used to read in analog signals from a peripheral
- Common analog peripherals include joysticks and variable resistors (like a dimmer dial for an LED)

02

Digital Pins

- Used to read/write digital signals to peripherals.
- For example you can set a digital pin to "HIGH" to light up an LED





IR Motor Control Demo

Feel Free to follow along if you have some components here!

01

Hardware

- Arduino Uno
- USB Cable
- IR Receiver
- IR Remote Control (any will do)
- Motor/Winch
- Appropriate Power Source For Your Motor/Winch

02

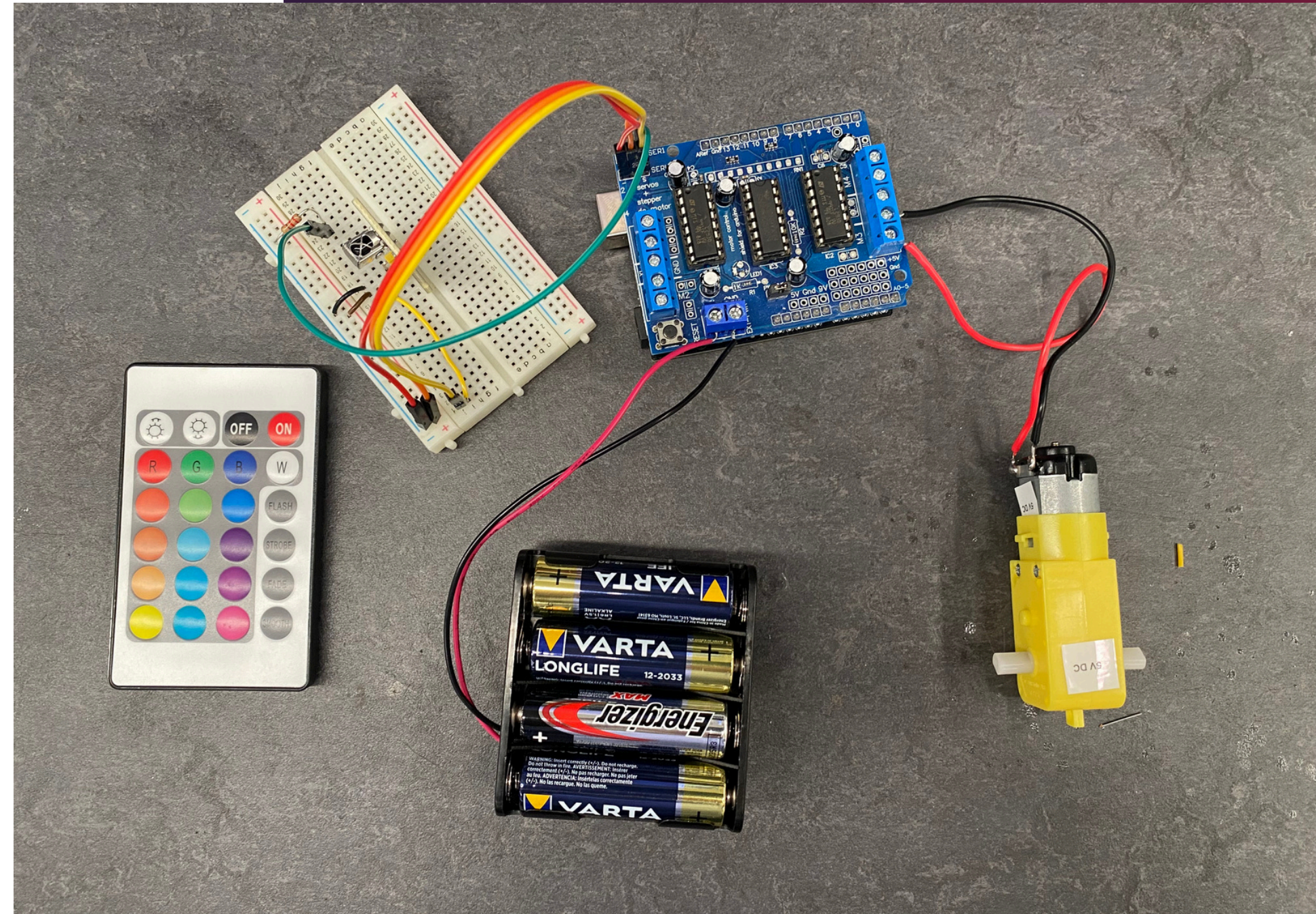
Software

- Arduino Programming IDE

03

Full Demo Code

- To make life easier, I placed all of the demonstration code on GitHub for your reference!
- All of the source code for this demonstration can be found on my GitHub account - <https://github.com/aburo8>





IR Motor Control Steps

01

Receiving IR Button Presses

Let's make our program receive the IR button presses.

02

Basic LED Light Control

Let's get a basic LED light blinking - we can use this for debugging later!

03

Turning on the Motor On/Off

Let's work out how to get the motor to turn off/on!

04

System Integration

Now that all the core system components are working separately! Lets save a copy of our code and they try and merge everything together so the IR remote toggles the motor.

05

Adding Extra Functionality

Instead of turning the motor on/off, why don't we try controlling it's speed?

06

Multi-Meter Checks

Let's check the voltage and current that our motor is drawing!

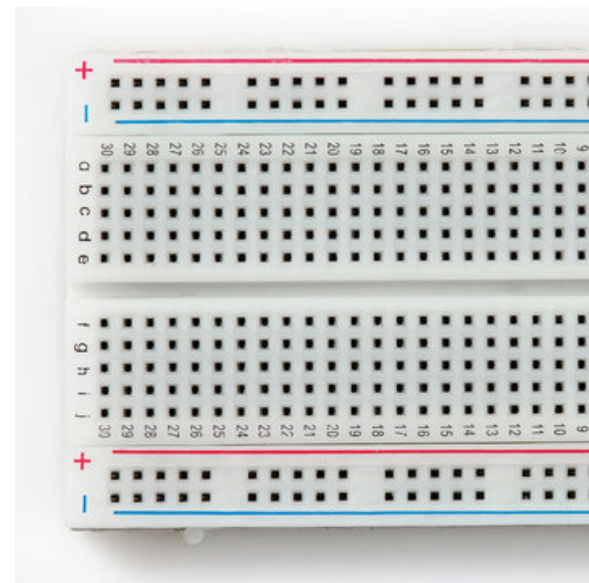


Full Demo Code

If you are struggling to follow along. You can see the full demo code here!

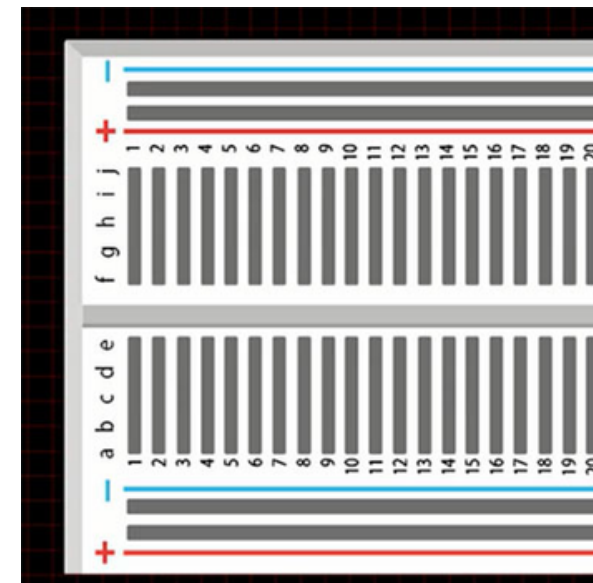


How Different Ways To Build Electronic Prototypes



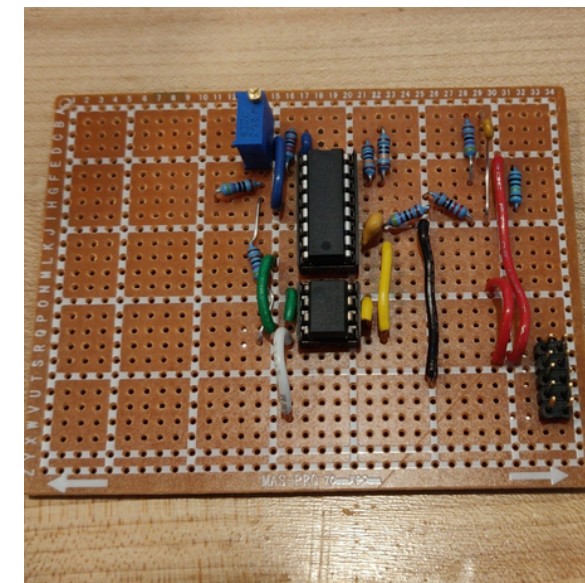
Breadboard

Breadboards are a fantastic way to get started with prototyping electronic projects! They are inexpensive and easy to use!



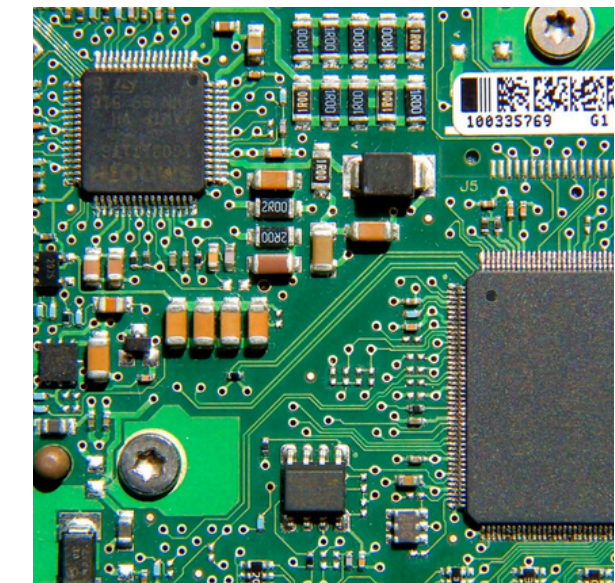
Breadboard Layout

This is the layout of a breadboard! The sections on the side are called “power rails” and are connected down the entirety of the breadboard. The columns in the middle are connected in rows.



Stripboard/Veroboard

A Stripboard is the next level up from a breadboard. You solder the components directly to the board so they can't move around or fall out of place. This is likely what you would want to do for your team project.



Printed Circuit Board (PCB)

After verifying your design on a Breadboard and/or Stripboard - you can fabricate a PCB. This is outside the scope of ENGG1100, but if you go into Electrical Engineering you will do this for Team Project II.



Tips & Tricks

01

Google Is Your Friend

If you are stuck with an Error, chances are someone else has run into in the past! Ask Google or ChatGPT for some troubleshooting advice.

02

Start with Code Examples

The Arduino Documentation comes with lots of code examples! If these aren't your style, you can always watch a YouTube video or ask ChatGPT for code examples too!

03

Use Existing Arduino Libraries

Where possible try and use existing Arduino libraries which have GOOD documentation! This will make your life a lot easier for debugging and troubleshooting.

04

SAVE SAVE SAVE!

Make backups/copies of working code after each major project milestone! This way you should always have a working copy of your project! This is a very important practice known as version-control.

05

Serial Monitor Debugging

Use the Arduino IDE Serial Monitor in conjunction with print statements to debug your code when things break!

06

Start Early!!!

If this is your first time writing code for an Arduino, make sure you start early. This gives you more time to solve and debug problems that you may run into.

Handy Resources

07

IR Control Demonstration Code



08

UQ Innovate Electronics Lab

Head to the UQ Innovate Electronics Lab if you need any assistance!



Q&A

We will wander around the room now and answer any 1-1 Questions!

We can also help you to get started! Or help to debug your code if you get stuck!

01

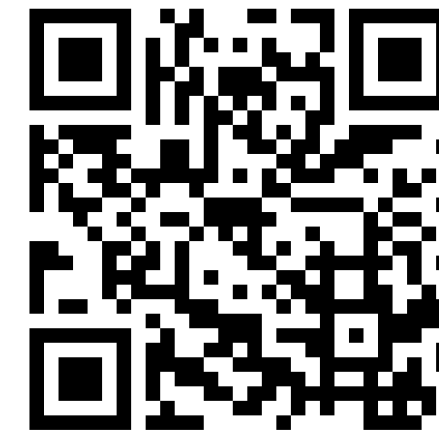
IEEE HKN Mu Kappa Chapter

Email: ieeekhkn.mukappa@gmail.com



02

IEEE Membership Signup



03

Demonstration Source Code





IEEE-Eta Kappa Nu (HKN)

Pizza Time!

All the best with your projects!

Please join us outside for some pizza, drinks & networking - brought to you by IEEE QLD Section!

01

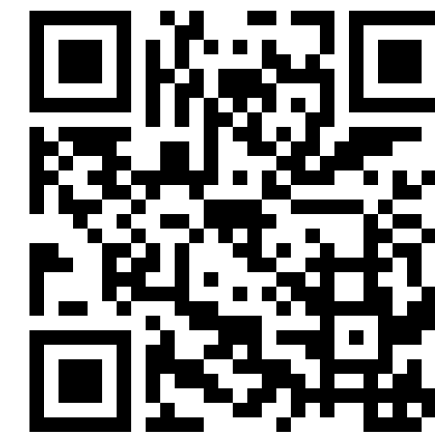
IEEE HKN Mu Kappa Chapter

Email: ieeehkn.mukappa@gmail.com



02

IEEE Membership Signup



03

Demonstration Source Code

